CS-663
Digital Image Processing (DIP)

Course-Project Examples

Suyash P. Awate

What is Expected from a Course Project

- Aim to use knowledge in class to explore beyond what is covered

1) Read interesting papers related to image analysis
2) Understand mathematics and algorithms underlying methods
3) Implement method(s); ideally, from scratch
4) Obtain/curate suitable real-world datasets for learning and/or testing
5) Comprehensive empirical analysis to study various aspects of how method works on various real-world scenarios. Ablation studies.
6) Summarize and explain results

- Present 1-6 in 10-mine presentation using PowerPoint/GoogleSlides
- Submit code + results + presentation on Moodle
- Grading on a relative scale

Image Enhancement/Restoration

- Instagram-filter effects on photos
- Virtual background (real time) in video calls
- Colorization of grayscale images & videos
  - e.g., ieeexplore.ieee.org/document/10892024
  - e.g., www.cs.huji.ac.il/~yweiss/Colorization/
  - e.g., richzhang.github.io/colorization/resources/colorful_eccv2016.pdf
- Red-eye effect removal
  - e.g., en.wikipedia.org/wiki/Red-eye_effect
- Video slo-mo
- Image style transfer (using CNNs)
  - e.g., www.kaggle.com/competitions/gan-getting-started/overview

Image Enhancement/Restoration

- Low-light image enhancement
  - e.g., openaccess.thecvf.com/content_CVPR_2020/papers/Guo_Zero-Reference_Deep_Curve_Estimation_for_Low-Light_Image_Enhancement_CVPR_2020_paper.pdf
  - e.g., daooshee.github.io/BMVC2018website/
- Image inpainting
  - e.g., Remove a photo-bomber
  - e.g., openaccess.thecvf.com/content_ECCV_2018/papers/Guilin_Liu_Image_Inpainting_for_ECCV_2018_paper.pdf
  - e.g., openaccess.thecvf.com/content/WACV2022/papers/Suvorov_Resolution-Robust_Large_Mask_Inpainting_With_Fourier_Convolutions_WACV_2022_paper.pdf

Image Enhancement/Restoration

- Artifact removal
  - e.g., mmlab.ie.cuhk.edu.hk/projects/ARCNN.html
  - e.g., dl.acm.org/doi/epdf/10.1145/1275808.1276390
  - Moire-pattern removal from photos of display screens
- Image/video resolution improvement
  - e.g., openaccess.thecvf.com/content_cvpr_2017/papers/Nah_Deep_Multi-Scale_Convolutional_CVPR_2017_paper.pdf
- AI-art generation:
  - e.g., Ghibli art generator
  - e.g., Pixel art generator
  - e.g., 1980s vintage art generator

Image Segmentation & Object Detection/Recognition

- Document scanning and automated reading
- License-plate detection, recognition
- Traffic-sign detection, recognition
- Instance segmentation
  - link.springer.com/chapter/10.1007/978-3-030-00934-2_30
- Object counting
  - e.g., count people in an image of a crowd
- Face recognition (FaceNet, DeepFace)
- Face expression analysis (using PCA or deep neural nets)
- Eye detection for red-eye removal

Image Segmentation & Object Detection/Recognition

- Crop-disease detection
- Leaf / flower / plant classification
- Iris recognition
- Anomaly detection in images
  - link.springer.com/chapter/10.1007/978-3-030-68799-1_35
- Skin-disease classification
- Hand-gesture recognition
- Tracking objects/features in a video

Image Understanding

- Remote photoplethysmography (rPPG)
  - e.g., www.sciencedirect.com/science/article/abs/pii/S1746809423010418

Datasets

- https://www.kaggle.com
- https://www.kaggle.com/datasets/?tags=13207-Computer+Vision
- https://grand-challenge.org
- https://www.image-net.org/
- https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html